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Internship : Data Analytics using Python
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Task 141 : Work with Big Data Tools (Hadoop, Spark) .

Objective

To gain hands-on experience in processing and analyzing large-scale datasets using distributed computing tools like Apache Hadoop and Apache Spark, enabling efficient data storage, faster computation, and scalable data analysis for real-world applications.

```
data = [
    ("Ayaan", "Laptop", 2, 50000),
    ("John", "Mobile", 3, 20000),
    ("Sara", "Laptop", 1, 50000),
    ("Ali", "Tablet", 2, 15000)
]

columns = ["Name", "Product", "Quantity", "Price"]

df = spark.createDataFrame(data, columns)


df.filter(df['Product']=='Laptop').show()
```

```
+-----+-------+--------+-----+
| Name|Product|Quantity|Price|
+-----+-------+--------+-----+
|Ayaan| Laptop|       2|50000|
| Sara| Laptop|       1|50000|
+-----+-------+--------+-----+
```

In this task, I worked with Apache Spark (PySpark) to perform a basic big data operation. I created a dataset containing customer purchase details such as Name, Product, Quantity, and Price, and then converted this data into a Spark DataFrame. After that, I applied a filter operation to extract only the records where the product is "Laptop". Finally, I displayed the filtered result using the .show() function, which returned only the rows related to laptop purchases. This task demonstrates how to use Spark for data filtering and basic data processing on structured datasets.

Task 142 : Use Statistical Software (SAS, SPSS) .

Objective:

To develop the ability to analyze and interpret data using statistical software such as IBM SPSS Statistics and SAS, enabling the application of statistical techniques like descriptive analysis, correlation, and regression to derive meaningful insights and support data-driven decision-making.

Variable View :

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
sepal_length	Numeric	3	1		None	None	8	Right	Scale	Input
sepal_width	Numeric	3	1		None	None	8	Right	Scale	Input
petal_length	Numeric	3	1		None	None	8	Right	Scale	Input
petal_width	Numeric	3	1		None	None	8	Right	Scale	Input
species	String	10	0		None	None	10	Left	Nominal	Input

Descriptive Statistics :

Descriptives

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
sepal_length	150	4.3	7.9	5.843	.8281
sepal_width	150	2.0	4.4	3.057	.4359
petal_length	150	1.0	6.9	3.758	1.7653
petal_width	150	.1	2.5	1.199	.7622
Valid N (listwise)	150				

In this task, I used IBM SPSS Statistics to perform descriptive statistical analysis on the dataset. First, I defined the variables in the Variable View, where I set attributes such as variable name, data type (numeric/string), and measurement scale (scale/nominal) for features like sepal length, sepal width, petal length, petal width, and species. After preparing the dataset, I moved to the analysis phase and applied Descriptive Statistics using the *Analyze* → *Descriptive Statistics* → *Descriptives* option. I calculated key statistical measures including mean, minimum, maximum, and standard deviation for each numeric variable.

Task 143 : Use Cloud-Based Data Tools (AWS, Google Cloud) .

Objective

To develop the ability to store, manage, and analyze data using cloud-based platforms such as Amazon Web Services and Google Cloud, enabling scalable, secure, and efficient data processing and real-time analytics for data-driven decision-making.

```
import pandas as pd

df = pd.read_csv('/kaggle/input/datasets/ayaan1011/user-data/User_Data_Association.csv')
df.head()
```

[1]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

In this task, I used a cloud-based platform (Kaggle) to perform data analysis, which simulates working with cloud tools like Amazon Web Services and Google Cloud.

I uploaded the dataset *User_Data_Association.csv* and loaded it into the environment using the pandas library in Python. I accessed the dataset using its cloud storage path and displayed the first few records using the .head() function to understand its structure.

The dataset contained fields such as User ID, Gender, Age, Estimated Salary, and Purchased.

This task helped me understand how data can be stored, accessed, and analyzed directly from a cloud-based environment without using local storage, demonstrating the basics of cloud data handling and analysis.

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype 
---  ------           --------------  ----- 
 0   User ID          400 non-null    int64 
 1   Gender           400 non-null    object
 2   Age              400 non-null    int64 
 3   EstimatedSalary  400 non-null    int64 
 4   Purchased        400 non-null    int64 
dtypes: int64(4), object(1)
memory usage: 15.8+ KB
```

In this step, I used the df.info() function in Python (pandas) to explore the structure of the dataset. This function provided a summary of the DataFrame, including the total number of entries (400 rows), the number of columns (5), and details about each column such as data type and non-null values.

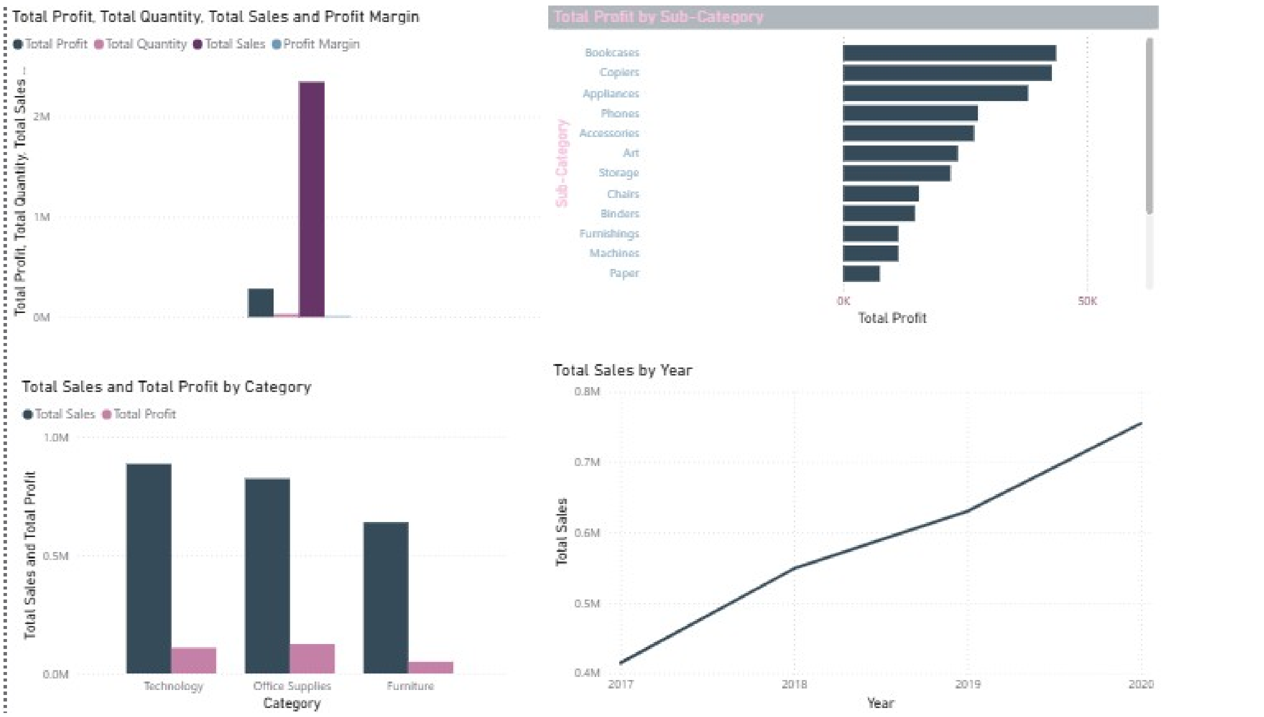
I observed that all columns (User ID, Gender, Age, Estimated Salary, and Purchased) contain complete data with no missing values. The data types include integers for numerical fields and object type for categorical data like Gender.

This step helped me understand the dataset structure, check data quality, and prepare it for further analysis in a cloud-based environment.

Task144: Use Tableau and Power BI for data visualization and reporting.

Objective

The main objective of using **Tableau** and **Power BI** is to transform raw data into **interactive, meaningful, and visually appealing dashboards and reports** that support better decision-making. These tools help users analyze trends, patterns, and insights quickly without complex coding.



In this task, I created an interactive dashboard in Power BI by visualizing sales data using multiple charts. I developed a bar chart to compare total sales and profit across different categories such as Technology, Office Supplies, and Furniture. I also created a horizontal bar chart to display total profit by sub-category, helping identify top-performing and low-performing products.

Additionally, I built a line chart to analyze total sales trends over the years, showing growth patterns from 2017 to 2020. I included key metrics like total profit, total quantity, total sales, and profit margin to provide a complete overview. These visualizations were combined into a single dashboard, making it easy to analyze performance and support data-driven decision-making.

Task 145 : Improve Communication and Presentation Skills .

Objective:

The objective of improving communication and presentation skills is to effectively convey ideas, insights, and data findings to team members and managers in a clear, confident, and professional manner. This includes the ability to explain complex information in a simple way, actively listen, and engage in meaningful discussions.

In this task, I presented my PowerPoint on the Data Analytics course at Lighthouse Foundation, where I explained key concepts and insights clearly to the audience.

I communicated the course content in a structured and simple manner, making it easy for others to understand.

During the presentation, I confidently spoke in front of the audience, answered questions, and engaged with listeners effectively.

This helped me improve my public speaking, confidence, and ability to present technical information in a clear and professional way.

Task 146 : Work on improving time management and efficiency .

Objective

The objective of this task is to develop effective time management skills and improve overall efficiency in completing tasks. It focuses on planning and prioritizing work, setting realistic deadlines, and using time wisely to achieve maximum productivity.

In this task, I improved my time management and efficiency by performing 10 tasks daily as part of my college OJT (On-the-Job Training).

I planned my daily schedule in advance, prioritized important tasks, and ensured that each activity was completed within the given time.

By following a structured routine and staying focused, I was able to manage multiple tasks effectively without delays.

This practice helped me enhance my productivity, maintain consistency in my work, and complete all assigned tasks efficiently.

Task 147 : Learn to choose the right data for analysis .

Objective

The objective of this task is to develop the ability to identify and select relevant and accurate data for analysis. It focuses on understanding which data is useful for solving a specific problem and removing unnecessary or irrelevant information.

```
import pandas as pd
```

```
df=pd.read_csv('data.csv')
print(df.head())
```

```
   Unnamed: 0  id   name  age     city  salary
0           0   1  Ayaan   25   Mumbai   50000
1           1   2   Sara   30     Pune   60000
2           2   3   John   28    Delhi   55000
3           3   4    Ali   22   Mumbai   45000
4           4   5  Meena   27  Chennai   52000
```

```
# Missing and Incorrect Values

print(df.isnull().sum())
```

```
Unnamed: 0    0
id            0
name          0
age           0
city          0
salary        0
dtype: int64
```

In this task, I worked on selecting the right data for analysis using Python and Pandas. I loaded the dataset from a CSV file and examined the structure using df.head() to understand the available columns such as id, name, age, city, and salary.

I identified an unnecessary column (Unnamed: 0) and recognized that it should be removed for better analysis. I also checked for missing or incorrect values using df.isnull().sum(), and confirmed that the dataset was clean with no null values.

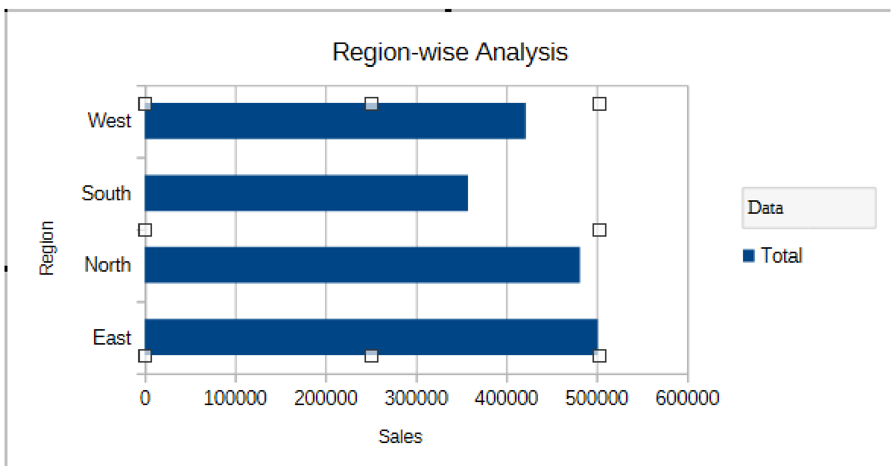
This process helped me ensure that only relevant, accurate, and meaningful data was used for further analysis, improving the quality of insights.

Task 148 : Provide useful data for decision-making .

Objective :

The objective of this task is to present meaningful, accurate, and relevant data that helps in making informed decisions. It focuses on analyzing data to identify patterns, trends, and key insights that support business or project goals.

Region	Sum - Sales
East	501428
North	481407
South	357687
West	421345
Total Result	**1761867**



Meaningful Insights:

- The **East region** has the **highest sales (~500,000)**, making it the top-performing region.
- The **North region** follows closely with around **480,000**, showing strong performance.
- The **West region** has moderate sales (~420,000).
- The **South region** records the **lowest sales (~350,000)**.

Task 149 :Participate in Soft Skills Training .

Objective :

The objective of this task is to improve essential interpersonal and professional skills required in a workplace environment. It focuses on developing communication, teamwork, problem-solving, leadership, and presentation skills.

Description :

In this task, I actively participated in soft skills training sessions where I improved my communication, teamwork, and presentation abilities. I engaged in group discussions, shared my ideas confidently, and collaborated with others to complete activities.

I also practiced public speaking by presenting topics in front of others, which helped me build confidence and reduce hesitation.

Through this training, I developed better interpersonal skills, learned how to work effectively in a team, and improved my overall professional behavior.

Task 150 :Evaluate and Improve Your Performance .

Objective :

The objective of this task is to assess your own performance and identify areas for improvement. It focuses on analyzing completed work, recognizing strengths and weaknesses, and taking corrective actions to enhance productivity and quality.

Description :

In this task, I evaluated and improved my performance by regularly reviewing my work and identifying areas where I could do better. I took feedback from mentors and worked on correcting my mistakes to enhance the quality of my work.

I also focused on learning new tools such as Power BI for data visualization, PySpark for big data processing, Excel Pivot Tables for data analysis, and IBM SPSS for statistical analysis.

By continuously practicing these tools and improving my skills, I was able to increase my efficiency, accuracy, and overall performance in data analytics tasks.